Exam :-

① $\int \frac{e^{\sin x}}{\sec x}\, dx$

Let $u = \sin x$, $du = \cos x\, dx$

$du = \frac{1}{\sec x}\, dx$

$$\int \frac{e^{\sin x}}{\sec x}\, dx = \int e^u\, du$$

$$= \frac{e^u}{u} + C$$

Sub $u = \sin x \Rightarrow \frac{e^{\sin x}}{\sin x} + C$

$$\therefore \boxed{\int \frac{e^{\sin x}}{\sec x}\, dx = \frac{e^{\sin x}}{\sin x} + C}$$

② $\int \sin^2 x \cos^3 x\, dx$

$$\Rightarrow \int \sin^2 x \cos^2 x \cos x\, dx$$

$$= \int \sin^2 x (1 - \sin^2 x) \cos x\, dx$$

Let $u = \sin x$, $du = \cos x\, dx$

$$= \int u^2 (1-u^2)\, du$$

$$= \int (u^2 - u^4)\, du$$

$$= \frac{u^3}{3} - \frac{u^5}{5} + C$$

Sub $u = \sin x$

$$\boxed{\int \sin^2 x \cos^3 x \, dx = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C}$$

(3) $$\int \frac{x+5}{x^2+4x+3}\, dx$$

$$= \int \frac{x}{x^2+4x+3}\, dx + \int \frac{5}{x^2+4x+3}\, dx$$

$$\int \frac{x}{x^2+4x+3}\, dx \Rightarrow \frac{A}{(x+1)} + \frac{B}{(x+3)}$$

$1+3 \overline{)\,3}\;4$
$1+3$

$\Rightarrow x = A(x+3) + B(x+1)$ Using partial faction.

Let $x = -3$

$-3 = -2B \Rightarrow \boxed{B = 3/2}$

Let $x = -1$

$-1 = 2A \Rightarrow \boxed{A = -1/2}$

$$\Rightarrow \int \frac{x}{x^2+4x+3}\,dx = -\int \left(\frac{1}{2}\right)\frac{1}{(x+1)}\,dx + \frac{3}{2}\int \frac{1}{(x+3)}\,dx$$

$$\Rightarrow \int \frac{x}{x^2+4x+3}\,dx = \frac{3}{2}\ln|x+3| - \frac{1}{2}\ln|x+1| + C$$

$$\int \frac{5}{x^2+4x+3}\,dx \Rightarrow \frac{A}{(x+1)} + \frac{B}{(x+3)}$$

using partial fraction

$$\Rightarrow 5 = A(x+3) + B(x+1)$$

Let $x = -3$

$$\Rightarrow 5 = -2B \Rightarrow \boxed{B = -5/2}$$

$x = -1 \Rightarrow 5 = 2A \Rightarrow \boxed{A = 5/2}$

$$\therefore \int \frac{5}{x^2+4x+3}\,dx = \frac{5}{2}\ln|x+1| - \frac{5}{2}\ln|x+3| + C$$

$$\therefore \int \frac{x+5}{x^2+4x+3}\,dx = \frac{3}{2}\ln|x+3| - \frac{5}{2}\ln|x+3| + 5/2\ln|x+1| - \frac{1}{2}\ln|x+1| + C$$

$$\boxed{\int \frac{x+5}{x^2+4x+3}\,dx = -\ln|x+3| + 2\ln|x+1| + C}$$

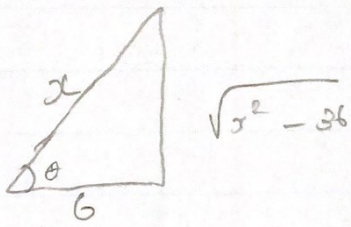
(4) $$\int \frac{2}{x^2\sqrt{x^2-36}}\,dx$$

$a = 6$

Let $x = a\sec\theta$

$x = 6\sec\theta$

$dx = 6\sec\theta\tan\theta\,d\theta$



$$= \int \frac{2}{36\sec^2\theta\sqrt{36\sec^2\theta-36}} \cdot 6\sec\theta\tan\theta\,d\theta$$

$$= \int \frac{\sec\theta\tan\theta\,d\theta}{3\sec^2\theta(6)\sqrt{\tan^2\theta}} = \int \frac{\tan\theta\,d\theta}{18\sec\theta\tan\theta}$$

$$= \int \frac{d\theta}{18\sec\theta} = \frac{1}{18}\int \cos\theta\,d\theta$$

$$= \frac{1}{18}\left[\sin\theta + C\right]$$

$$= \frac{1}{18}\left[\frac{\sqrt{x^2-36}}{x}\right] + C$$

$$\int \frac{2}{x^2\sqrt{x^2-36}}\,dx = \frac{\sqrt{x^2-36}}{18x} + C$$

(5) $\int_2^4 \sqrt{x}\, \ln(x)\, dx$

Let $u = \ln(x)$ | $dv = \sqrt{x}\, dx$

$du = \frac{1}{x} dx$ | $v = \frac{2}{3} x^{3/2} + C$

$\int u\,dv = uv - \int v\,du$

$$\int \sqrt{x} \ln x\, dx = \frac{2}{3} \ln(x)\, x^{3/2} - \int \frac{2}{3} x^{3/2} \frac{1}{x} dx$$

$$= \frac{2}{3} \ln(x)\, x^{3/2} - \frac{2}{3} \int x^{1/2} dx$$

$$= \frac{2}{3} \ln(x)\, x^{3/2} - \frac{2}{3} \frac{x^{3/2}}{3/2} + C$$

$$= \frac{2}{3} \ln(x)\, x^{3/2} - \frac{4}{9} x^{3/2} + C$$

$$\int_2^4 \sqrt{x} \ln(x)\, dx = \left[\frac{2}{3} \ln(x)\, x^{3/2} - \frac{4}{9} x^{3/2} \right]_2^4$$

$$= \left[\frac{2}{3} \ln(4)\, (4)^{3/2} - \frac{4}{9} (4)^{3/2} \right] - \left[\frac{2}{3} \ln(2)\, (2)^{3/2} - \frac{4}{9} (2)^{3/2} \right]$$

$$= \frac{16 \ln(4)}{3} - \frac{32}{9} - \frac{4\sqrt{2} \ln(2)}{3} + \frac{8\sqrt{2}}{9}$$

$$\int_2^4 \sqrt{x}\,\ln(x)\,dx = \frac{16\ln(4) - 4\sqrt{2}\ln(2)}{3} + \frac{8\sqrt{2} - 32}{9}$$

(6) $$\int_1^e \frac{1}{x\sqrt{\ln x}}\,dx$$

$$\int \frac{1}{x\sqrt{\ln x}}\,dx \Rightarrow \text{Let } u = \ln x \;,\; du = \frac{1}{x}\,dx$$

$$= \int \frac{1}{\sqrt{u}}\,du = \int u^{-1/2}\,du$$

$$= \frac{u^{1/2}}{1/2} + C = 2u^{1/2} + C$$

$u = \ln(x) \Rightarrow$

$$\int \frac{1}{x\sqrt{\ln(x)}}\,dx = 2\sqrt{\ln(x)} + C$$

$$\int_1^e \frac{1}{x\sqrt{\ln(x)}}\,dx = \left[2\sqrt{\ln(x)}\right]_1^e$$

$$= \left[2\sqrt{\ln e^1} - 2\sqrt{\ln(1)}\right]$$

$= 2$ but when $\frac{1}{2\sqrt{\ln(x)}}$ is improper intergal at $\boxed{x=1}$.

$$\therefore \int_1^e \frac{1}{x\sqrt{\ln(x)}}\,dx = 2$$

(7)

(A) $\sum_{n=1}^{\infty} \frac{n^2 - 4n}{3n^4+9}$

Using comparison test :-

$\sum_{n=1}^{\infty} \frac{n^2-4n}{3n^4+9}$ compare to $\sum_{n=1}^{\infty} \frac{n^2}{3n^4}$

$\sum_{n=1}^{\infty} \frac{n^2}{3n^4} = \sum_{n=1}^{\infty} \frac{1}{3n^2}$ which is converges by P-series

$P = 2 > 1$

$\sum_{n=1}^{\infty} \frac{n^2-4n}{3n^4+9} < \sum_{n=1}^{\infty} \frac{n^2}{3n^4}$ because

$n^2 - 4n < n^2$ and $3n^4 + 9 > 3n^4$.

$\therefore \sum_{n=1}^{\infty} \frac{n^2-4n}{3n^4+9}$ is converges by comparison test.

(B) $\sum_{n=1}^{\infty} \frac{(2n)!}{3^n (n!)^2} = \sum_{n=1}^{\infty} \frac{(2n)!}{3^n (n!)(n!)}$

Using Ratio test :-

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{[2(n+1)]!}{3^{n+1}\,(n+1)!\,(n+1)!} \cdot \frac{3^n (n!)(n!)}{(2n)!} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(2n+2)(2n+1)\cancel{(2n)!}\,\cancel{(3^n)}\,\cancel{(n!)}\,\cancel{(n!)}}{\cancel{(2n)!}\,\cancel{(3^n)}(3)\,(n+1)\,\cancel{(n!)}\,(n+1)\,\cancel{(n!)}} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(2n+2)(2n+1)}{3(n+1)(n+1)} \right| = \lim_{n\to\infty} \left| \frac{4n^2+4n+2n+2}{3(n^2+2n+1)} \right|$$

$$= \lim_{n\to\infty} \left| \frac{4n^2+6n+2}{3n^2+6n+3} \right| = \frac{4}{3} > 1$$

$\therefore L = 4/3 > 1$

$\therefore$ $\sum_{n=1}^{\infty} \frac{(2n)!}{3^n (n!)^2}$ is diverges by Ratio test

(8) $\sum_{n=1}^{\infty} (-1)^n \cdot n e^{-n^2} = \sum_{n=1}^{\infty} \frac{(-1)^n n}{e^{n^2}}$

Since Alternating ! we can use A.S.T

Alternating Series test :-

with $a_n = \frac{n}{e^{n^2}}$

① is positive on $[1,\infty)$? yes, because

$\frac{n}{e^{n^2}} = \frac{\text{positive}}{\text{positive}}$ ∴ positive for all positive 'n'. on $[1,\infty)$.

② is decreasing on $[1,\infty)$?

$\frac{n+1}{e^{(n+1)^2}} < \frac{n}{e^{n^2}}$ ∴ yes, a_n is decreasing.

③ $\lim_{n\to\infty} a_n = 0$? $\lim_{n\to\infty} \frac{n}{e^{n^2}} = 0$ yes. by applying limit.

∴ $\sum_{n=1}^{\infty} (-1)^n n e^{-n^2}$ is Converges absolutely.

(9)

(A) $\sum_{n=1}^{\infty} \frac{x^n}{n \cdot 3^n}$

Using Ratio test :-

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{(n+1)(3^{n+1})} \cdot \frac{n 3^n}{x^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{x \, x^n \cdot n \cdot 3^n}{(n+1)(3)(3^n) \cdot x^n} \right| = \lim_{n\to\infty} \left| \frac{x\, n}{3(n+1)} \right|$$

$$= \lim_{n\to\infty} \left| \frac{x}{3} \right| \frac{n}{(n+1)} = \left| \frac{x}{3} \right| < 1$$

$\therefore -1 < \frac{x}{3} < 1 \quad \Rightarrow \quad -3 < x < 3$

Check end points :-

when $x = -3$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-3)^n}{n\, 3^n} = \sum_{n=1}^{\infty} (-1)^n \left(\frac{3^n}{3^n} \right) \frac{1}{n}$$

Converges by A.S.T

when $x = 3$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{3^n}{3^n} \frac{1}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$$

$\therefore$ Diverges by limit

∴ Interval of convergence = $[-3, 3)$ → (1)

Radius = 3 → (2)

(3) $\sum_{n=1}^{\infty} \frac{x^n}{n^2 \cdot 3^n}$

Using Ratio test :-

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{(n+1)^2 3^{n+1}} \cdot \frac{3^n n^2}{x^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{x}{3} \right| \frac{n^2}{n^2+1} = \left| \frac{x}{3} \right|$$

$$L = \left| \frac{x}{3} \right| < 1$$

$$\therefore -1 < \frac{x}{3} < 1 \Rightarrow -3 < x < 3$$

Check end points :-

when $x = -3$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n^2 \cdot 3^n} = \sum_{n=1}^{\infty} (-1)^n \left(\frac{3}{3^n}\right) \frac{1}{n^2} = 0$$

↓

Converges by A.S.T

when $x = 3$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{3^n}{3^n} \frac{1}{n^2} \Rightarrow \text{Converges by P-series}$$

$P = 2 > 1$

∴ Interval of convergence = $[-3, 3]$ → (3)

Radius = 3 → (4)

From (2) and (4)

Radius of convergence for $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ and

$\sum_{n=1}^{\infty} \frac{x^n}{n^2 3^n}$ are Same.

But from (1) and (3)

the Interval of convergence for

$\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ and $\sum_{n=1}^{\infty} \frac{x^n}{n^2 3^n}$ are different on end points

(10) $f(x) = \frac{1}{x}$ centered at $a = 3$

$$f(x) = f(a) + f'(a)(x-a) + f''(a)\frac{(x-a)^2}{2!} + \cdots + f^n(a)\frac{(x-a)^n}{n!}$$

$f(a) = 1/3$

$f'(x) = (-1)x^{-2} \Rightarrow -1/9$

$f''(x) = (-1)(-2)x^{-3} \Rightarrow 2/27$

$\vdots$

$f^n(x) = (-1)^n (n!) x^{(-n-1)} \Rightarrow (-1)^n \frac{(n!)}{3^{n+1}}$

$$\therefore f(x) = \frac{1}{3} + \left(-\frac{1}{9}\right)(x-3) + \left(\frac{2}{27}\right)\frac{(x-3)^2}{2} + \cdots$$

$$+ \frac{(-1)^n (n!)}{3^{n+1}} \cdot \frac{(x-3)^n}{(n!)} + \cdots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (n!)\ (x-3)^n}{3^{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n (x-3)^n}{3^{n+1}}$$

$$\therefore \boxed{f(x) = \frac{1}{x} = \sum_{n=0}^{\infty} \frac{(-1)^n\ (x-3)^n}{3^{n+1}}}$$

(12) $f(x) = x \ln(x^3+1)$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n \quad \rightarrow \quad \int \frac{1}{1+x}\,dx = \ln|1+x| + C \Rightarrow \int (-1)^n x^n\,dx$$

$$= (-1)^n \frac{x^{n+1}}{n+1}$$

$$\therefore \ln(x+1) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} \qquad \text{replace } x \text{ with } x^3$$

$$\therefore \ln(x^3+1) = \sum_{n=0}^{\infty} (-1)^n \frac{(x^3)^{n+1}}{n+1} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{n+1}$$

$$f(x) = x \ln(x^3+1)$$

$$= x \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+3}}{n+1}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+4}}{n+1}$$

$$\therefore f(x) = x \ln(x^3+1) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+4}}{n+1}$$

First four non zero terms :-

when: $n=0 \Rightarrow \frac{x^4}{1}$

$n=1 \Rightarrow (-1)\frac{x^7}{2}$

$n=2 \Rightarrow \frac{x^{10}}{3}$

$n=3 \Rightarrow (-1)\frac{x^{13}}{4}$

$$\therefore \boxed{f(x) = x^2 - \frac{x^7}{2} + \frac{x^{10}}{3} - \frac{x^{13}}{4}}$$

(12) $\int_0^{1/2} \cos x \, dx$

(A)

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^{2k} \frac{x^{2k}}{(2k)!}$$

$$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

$$\int_0^{1/2} \cos x \, dx \approx \int_0^{1/2} \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}\right) dx$$

$$\approx \left[x - \frac{x^3}{3(2!)} + \frac{x^5}{5(4!)} - \frac{x^7}{7(6!)}\right]_0^{1/2}$$

$$\approx \left[\frac{1}{2} - \frac{(1/2)^3}{6} + \frac{(1/2)^5}{120} - \frac{(1/2)^7}{5040}\right] - [0]$$

$$\approx \frac{1}{2} - \frac{1}{48} + \frac{1}{3840} - \frac{1}{645120}$$

$$T_6(x) \approx 0.479425332$$

(B)

$$f(x) = T_6 + (R_6)$$

$$\therefore \quad |R_6| \leq \int_0^{1/2} \frac{x^8}{(8!)}\, dx$$

$$= \frac{x^9}{(9)(8!)}\Big|_0^{1/2}$$

$$= \frac{(1/2)^9}{(9)(8!)} \quad -(8)$$

$$= \frac{1}{185794560}$$

$$\boxed{|R_6| \leq 0.0000000538}$$